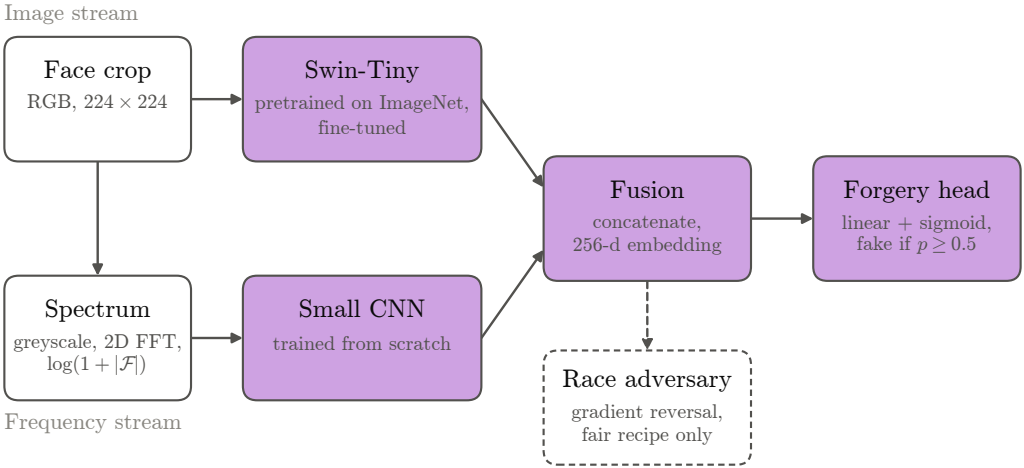
Image stream
Face crop
RGB, 224 × 224
Swin-Tiny
pretrained on ImageNet, fine-tuned
Spectrum
greyscale, 2D FFT, log(1 + |F|)
Small CNN
trained from scratch
Frequency stream
Fusion
concatenate, 256-d embedding
Forgery head
linear + sigmoid, fake if p ≥ 0.5
Race adversary
gradient reversal, fair recipe only